SAMSUNG | UNDP

Innovation Campus

Global Hunger Index 2025 Prediction

A Machine Learning-Based Approach to Predicting Global Hunger Levels

This project focuses on predicting 2025 Global Hunger Index scores by using machine learning techniques and World Bank indicators such as sanitation services and forest area.

The main aim is to create a more reliable prediction model by moving beyond historical GHI scores and integrating real-world socio-economic and environmental variables.

SDG Focus:

United Nations Sustainable Development Goal 2: Zero Hunger

Together for Tomorrow!
Enabling People
Education for Future Generations

Background

- Global hunger remains one of the most critical global challenges, affecting food security, nutrition, public health and long-term sustainable development.
- The Global Hunger Index is an important tool used to measure and compare hunger levels across countries. It helps identify countries facing serious food insecurity and nutrition-related challenges.
- In the previous phase of the project, 2025 GHI scores were predicted using a Linear Regression model based on historical GHI values from 2000, 2008 and 2016.
- Although the initial model produced strong results, it mainly relied on past GHI scores. Therefore, the project was refined to build a more realistic model based on external development indicators.

Objectives

- The main objective of this project is to develop a machine learning model that predicts 2025 Global Hunger Index scores more reliably.
- The project aims to:
- Predict 2025 GHI scores using machine learning methods
- Reduce data leakage by removing historical GHI scores from input features
- Integrate real-world indicators from the World Bank API
- Compare advanced models such as Random Forest and XGBoost
- Select the most reliable model for final prediction and deployment
- Deploy the final model through a FastAPI-based cloud API
- Support SDG 2: Zero Hunger through a data-driven analytical approach

SDG Relation

This project is directly related to the United Nations Sustainable Development Goal 2: Zero Hunger.

SDG 2 aims to end hunger, achieve food security, improve nutrition and promote sustainable agriculture. Since our project focuses on predicting Global Hunger Index scores, it directly supports the analysis of hunger and food insecurity risks.

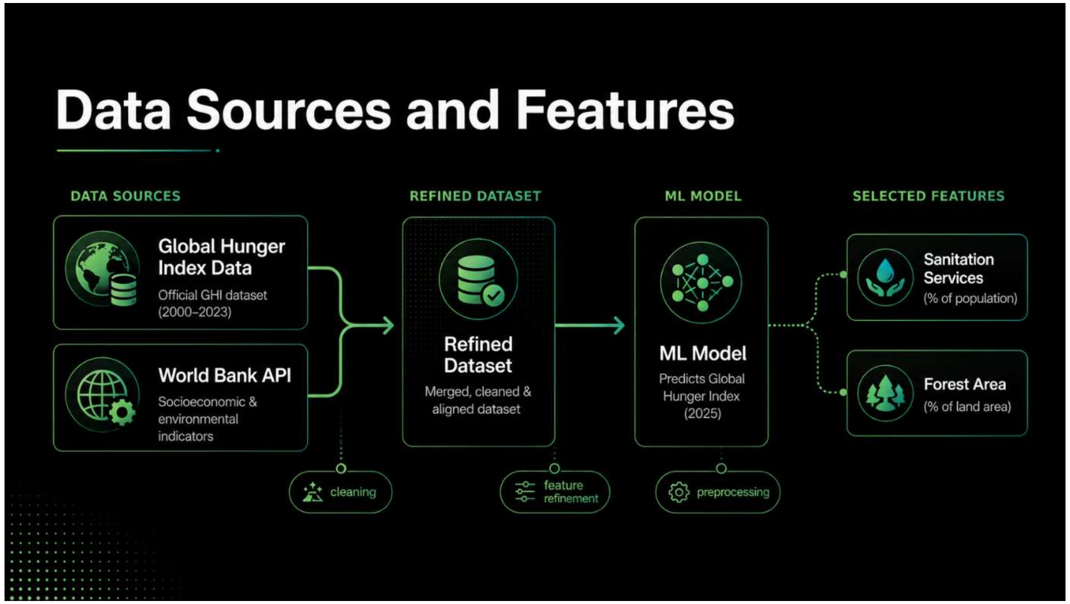
The refined model uses external indicators such as sanitation services and forest area instead of relying only on historical hunger scores.

Sanitation services are related to public health, poverty reduction, infrastructure quality and nutrition outcomes. Forest area is related to environmental sustainability, agricultural capacity and ecological stability.

By using these indicators, the project provides a more meaningful and data-driven perspective on the structural factors behind hunger.

Data

- The project uses Global Hunger Index data and additional development indicators collected from the World Bank API.
- In the previous model, historical GHI scores from 2000, 2008 and 2016 were used to predict 2025 GHI scores. However, these historical target-related variables were removed from the refined model to avoid data leakage.
- New external features were added to improve the model:
- Sanitation Services (% of population)
- Forest Area (% of land area)
- During testing, the independent variables were aligned with the same feature framework used in model development. Incomplete observations were removed, and the test features were standardized using the scaling parameters learned from the training data.
- The refined dataset allows the model to learn from real socio-economic and environmental indicators instead of only following historical GHI patterns.

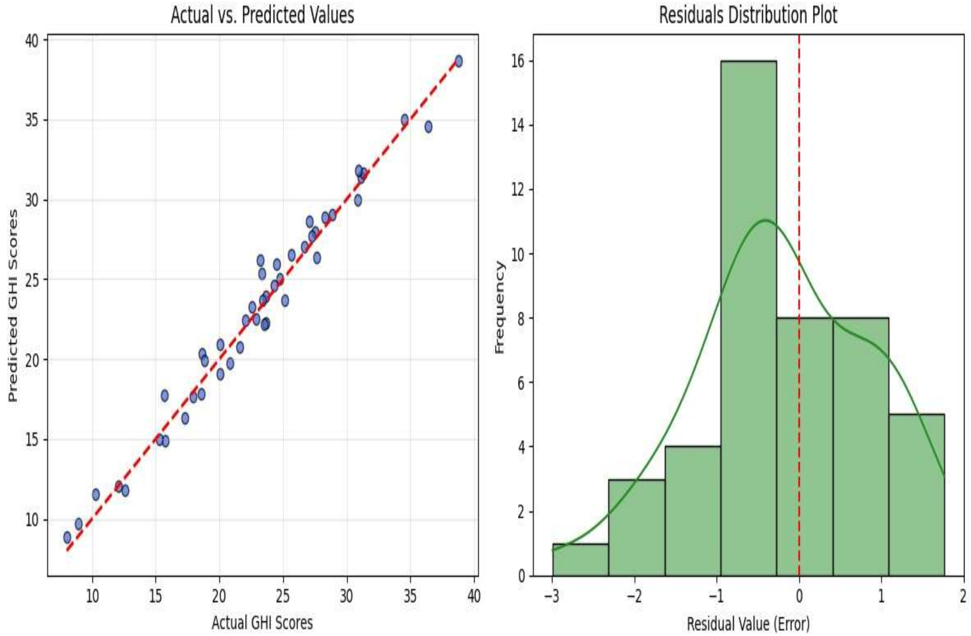


Model Selection and Training/Testing

- In the refined model pipeline, historical GHI scores were removed from the input features to avoid data leakage.
- Two advanced machine learning models were tested:
- Random Forest Regressor
- XGBoost Regressor
- Hyperparameter tuning was performed using GridSearchCV. The models were also evaluated with 5-Fold Cross-Validation to test their ability to generalize to unseen data.
- The optimized Random Forest Regressor achieved a lower cross-validation RMSE than XGBoost:
- Random Forest 5-Fold CV RMSE: 9.4886
- XGBoost 5-Fold CV RMSE: 9.6963
- Therefore, the optimized Random Forest Regressor was selected as the final model for test-set prediction and deployment.

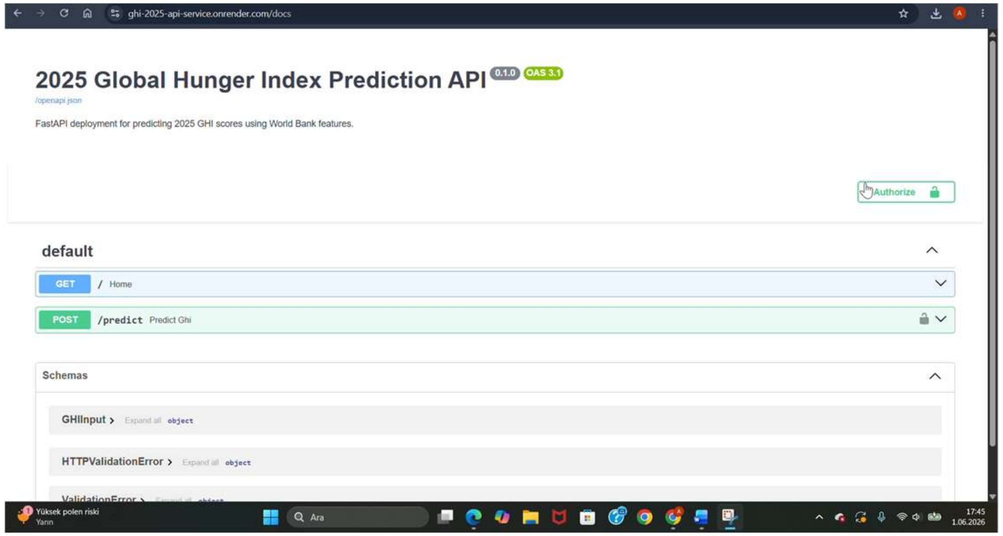
Evaluation Results

- The optimized Random Forest Regressor was evaluated on an independent test dataset representing 20% of the cleaned global observations.
- Final test performance:
- R^2 Score: 0.8415
- MAE: 2.5210
- RMSE: 3.1042
- These results show that the model explains approximately 84.15% of the variance in 2025 GHI scores using only sanitation services and forest area indicators.
- Compared with the previous Linear Regression model, the refined model has slightly lower raw performance metrics. However, the previous model relied on historical GHI scores, which created data leakage.
- Therefore, the refined model is methodologically more reliable because it predicts hunger levels using external development indicators rather than past target values.
- The Actual vs. Predicted plot shows that predictions are closely aligned with actual GHI scores, while the residual distribution supports a balanced and deployable predictive structure.



Deployment

- The final optimized Random Forest model was serialized as:
- **best_ghi_model.pkl**
- The model was deployed through a FastAPI-based RESTful API on the Render cloud platform.
- API endpoints:
- GET / : Health-check route
- POST /predict : Main prediction route
- The /predict endpoint receives two input variables in JSON format:
- Sanitation_Services_Pct_Pop
- Forest_Area_Pct_Land
- The API returns:
- Operational status
- Predicted 2025 GHI score
- Received input parameters
- Execution time
- Security was implemented using API key authentication through the X-API-KEY header. Logging was also added to record request time, input values, predicted GHI score and execution latency.



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Thank you!